

Azure Data Factory Real Time Scenario Part - 29

Copy Multiple Tables in Bulk From Azure SQL Database to ADLS

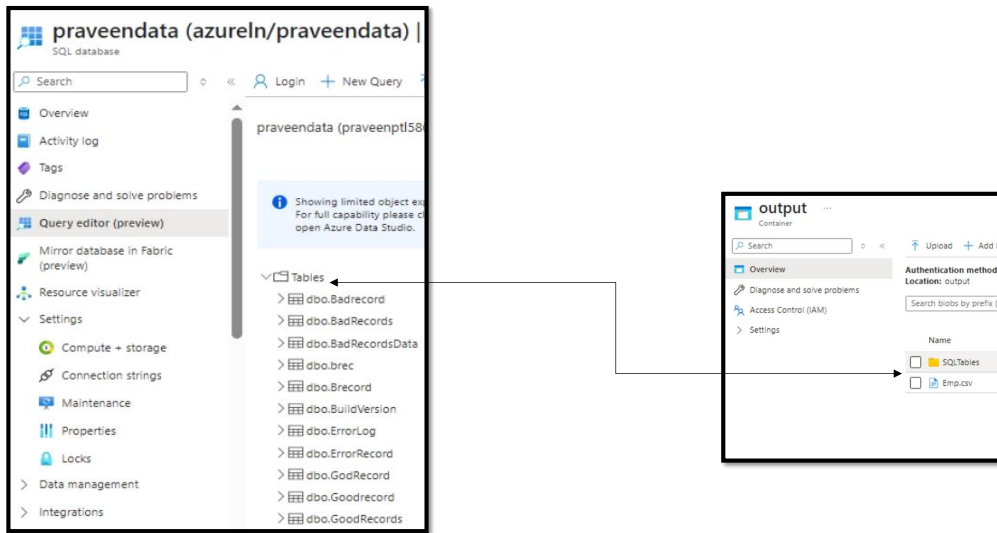
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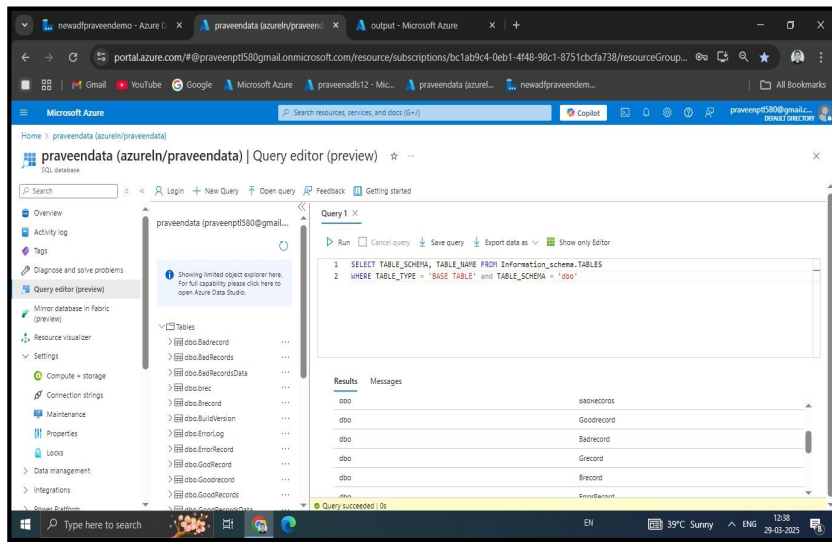
Business Requirement

Suppose i have multiple tables with different schema in Azure SQL database as per the requirement i have to copy those tables from database to Azure Data Lake Storage whose schema is dbo.



Query to Access All the Tables that Present in dbo Schema

When you open Azure SQL Database query editor to write the following query to execute then it will give you all the tables that present in specific schema.



The screenshot displays the Azure SQL Database query editor interface. The query editor shows the following SQL query:

```
1 SELECT TABLE_SCHEMA, TABLE_NAME FROM Information_schema.TABLES
2 WHERE TABLE_TYPE = 'BASE TABLE' and TABLE_SCHEMA = 'dbo'
```

The query results are displayed in a table with two columns: TABLE_SCHEMA and TABLE_NAME. The results show the following tables:

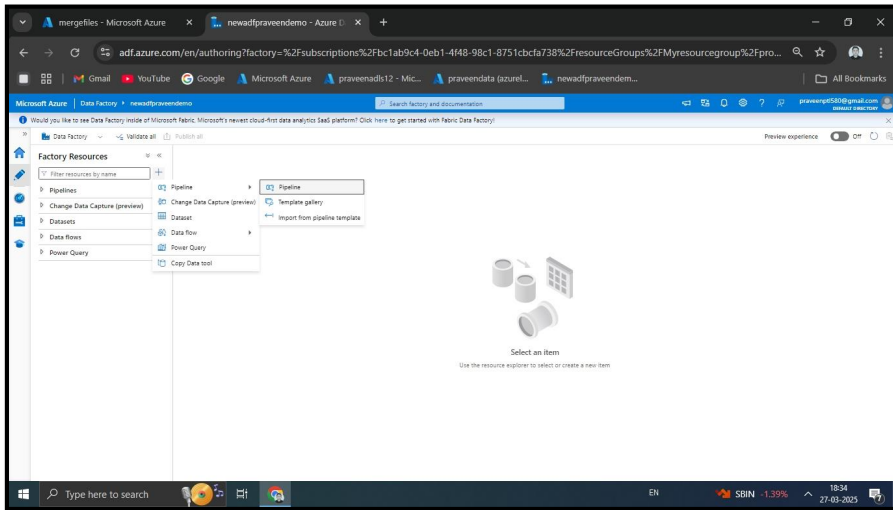
TABLE_SCHEMA	TABLE_NAME
dbo	BadRecord
dbo	GoodRecord
dbo	BadRecord
dbo	GoodRecord
dbo	BadRecord
dbo	GoodRecord
dbo	BadRecord
dbo	GoodRecord
dbo	BadRecord
dbo	GoodRecord

The status bar at the bottom indicates that the query was successful and executed in 0 seconds.

Practical Demonstration

Step 1

Open Azure Data Factory then go to author tab click on '+' icon and create a new pipeline.



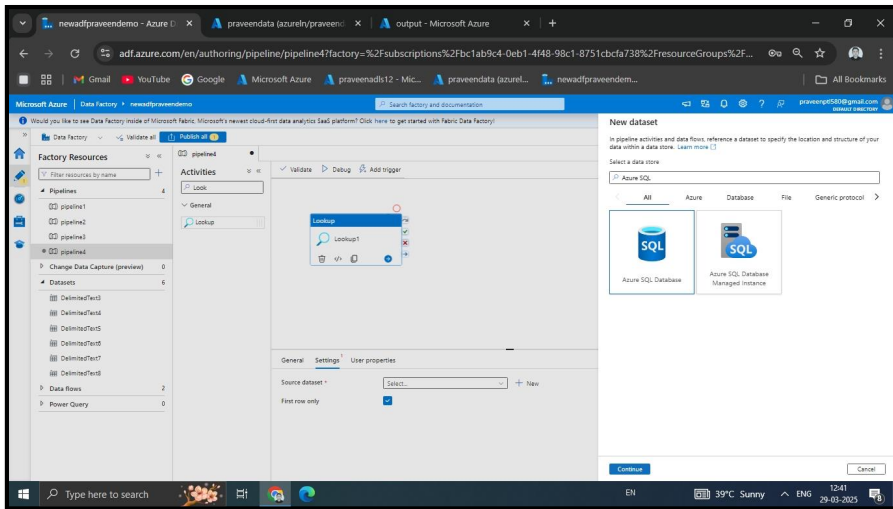
Step 2

Go to activity search bar search get Lookup activity then select, drag and drop it.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline4' selected. The main workspace is titled 'pipeline4' and contains a search bar with the word 'Lookup' entered. Below the search bar, a 'Lookup' activity icon is visible. The right-hand 'Properties' pane is open, showing the configuration for the selected 'Lookup1' activity. The 'General' tab is active, displaying fields for 'Name' (set to 'Lookup1'), 'Description', 'Activity state' (set to 'Activated'), 'Timeout' (set to '0:12:00:00'), and 'Retry' (set to '0'). The bottom of the screen shows the Windows taskbar with the search bar and system clock indicating 12:40 on 29-03-2025.

Step 3

Next go to settings click on pencil icon to create a new source dataset then search and select a new dataset for Azure SQL database then click on continue button.



Step 4

In set properties make a new dataset name select linked service for Azure SQL database dont select any table next click on ok button.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists pipelines (pipeline1, pipeline2, pipeline3, pipeline4) and datasets (DelimitedText3 through DelimitedText8). The 'Activities' pane shows a 'Lookup' activity selected. The main canvas displays a 'Lookup' activity icon. The 'Set properties' dialog is open on the right, with the following fields: 'Name' (AzureSqlTable1), 'Linked service' (AzureSqlDatabase1), 'Table name' (Select...), 'Import schema' (None), and 'Source dataset' (Select...). The 'First row only' checkbox is checked. The 'OK' button is highlighted.

Step 5

Next disable first row only then select use query as Query write the following query in the Query field to retrieve all the tables with dbo schema.

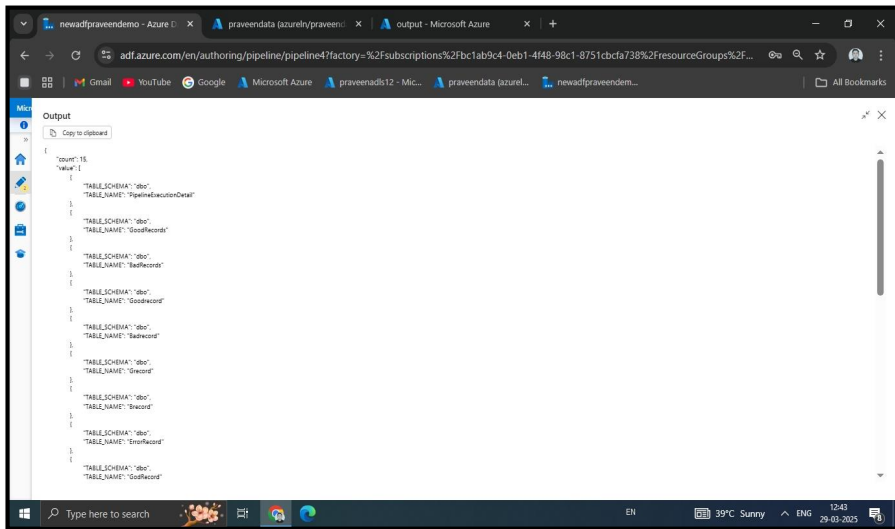
The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' expanded, listing 'pipeline1' through 'pipeline4'. 'pipeline4' is selected. Below it, 'Datasets' are listed, including 'AzureSqlTable1' and several 'DelimitedText' datasets. The main workspace shows the 'Lookup' activity configuration for 'pipeline4'. The 'Activities' pane on the left has 'Lookup' selected. The 'Lookup' activity is shown in a visual editor with a 'Lookup1' activity icon. Below the visual editor, the 'Settings' tab is active, showing the 'Use query' option selected under 'User properties'. The 'Query' field contains the following SQL query:

```
SELECT TABLE_SCHEMA, TABLE_NAME FROM information_schema.TABLES  
WHERE TABLE_TYPE = 'BASE TABLE' and TABLE_SCHEMA = 'dbo'
```

 The 'Properties' pane on the right shows the 'General' tab with the 'Name' field set to 'Lookup1'. The bottom status bar indicates the user is logged in as 'praveenpatel580@gmail.com' and the system time is 12:42 on 29-03-2025.

Step 6

Now execute Lookup Activity using debug mode when it executes successfully you get to see all the table with schema dbo in json code value property.



The screenshot shows a web browser window with the URL `adf.azure.com/en/authoring/pipeline/pipeline4?factory=%2Fsubscriptions%2Fbc1ab9c4-0eb1-4f48-98c1-8751bcfa738%2FresourceGroups%2F...`. The browser tabs include 'newadpravedemo - Azure D...', 'praveendata (azureln/praveend...', and 'output - Microsoft Azure'. The 'Output' window is open, displaying a JSON array of objects. Each object contains a 'count' of 15 and a 'value' array of objects. Each object in the 'value' array has 'TABLE_SCHEMA' set to 'dbo' and 'TABLE_NAME' set to a specific table name. The table names are: 'PipelineExecutionDetail', 'GoodRecords', 'BadRecords', 'GoodRecord', 'BadRecord', 'Grecord', 'Brecord', and 'GoodRecord'.

```
{
  "count": 15,
  "value": [
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "PipelineExecutionDetail"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "GoodRecords"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "BadRecords"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "GoodRecord"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "BadRecord"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "Grecord"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "Brecord"
    },
    {
      "TABLE_SCHEMA": "dbo",
      "TABLE_NAME": "GoodRecord"
    }
  ]
}
```

Step 7

Go to activity search bar search for Each activity then select drag and drop it next add lookup activity to forEach activity with green arrow.

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the user is logged in as 'praveenge590@gmail.com'. The main workspace is titled 'pipeline4' and shows a visual representation of the pipeline. The pipeline consists of two activities: a 'Lookup' activity (labeled 'Lookup1') and a 'ForEach' activity (labeled 'ForEach1'). A green arrow indicates the flow from the 'Lookup' activity to the 'ForEach' activity. The 'Activities' pane on the left lists the available activities, including 'ForEach' and 'Lookup'. The 'Properties' pane on the right shows the details for the selected 'ForEach' activity, including its name 'ForEach1' and its state 'Activated'. The 'General' tab is selected, showing the activity's name, description, and state. The 'Settings' tab is also visible, showing the activity's configuration. The 'Activities' tab is currently empty, indicating that no activities have been added to the 'ForEach' loop yet.

Step 8

Next go to settings enable sequential then go to Items add dynamic content then go to activity outputs select lookup1 activity outputs write 'value' after ' '. For each activity iterates each value that contains table_schema and table_name next click on ok button.

The screenshot displays the Microsoft Azure Data Factory Pipeline editor. The left sidebar shows the 'Factory Resources' tree with 'pipeline4' selected. The main canvas shows a pipeline diagram with a 'Lookup' activity followed by a 'ForEach' activity. The 'Settings' tab for the 'ForEach' activity is open, showing the 'Sequential' checkbox checked and the 'Items' property set to 'Add dynamic content (Alt+Shift+Q)'. The 'Pipeline expression builder' on the right shows the expression `@activity('Lookup1').output.value`. Below the expression builder, the 'Activity outputs' tab is selected, showing the output of the 'Lookup1' activity, which includes 'Lookup1 activity output', 'Lookup1 count', 'Count of the rows', and 'Lookup1 value array'.

Step 9

Next go to for each add activities and add copy activity.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline4?factory=%2Fsubscriptions%2Fbc1ab9c4-0eb1-4f48-98c1-8751bcfa738%2FresourceGroups%2F...`. The left sidebar shows the 'Factory Resources' tree with 'pipelines' expanded, listing 'pipeline1', 'pipeline2', 'pipeline3', and 'pipeline4'. The main canvas shows the 'pipeline4' diagram with a 'Lookup' activity connected to a 'ForEach' activity. The 'ForEach' activity is currently selected, and its context menu is open, showing options like 'Search', 'Move and transform', 'Copy data', 'Data flow', 'Synapse', 'Notebook', 'Spark job definition', 'Azure Data Explorer', 'Azure Data Explorer Command', 'Azure Function', 'Batch Service', 'Custom', 'Databricks', and 'Notebook'. The 'Properties' pane on the right shows the 'General' tab for the selected activity, with 'Name' set to 'pipeline4' and a description field. The bottom status bar indicates the user is logged in as 'praveen@580@gmail.com' and the system time is 12:46 on 29-03-2025.

Step 10

Now go to copy activity source next create a new source dataset search and select new dataset for Azure SQL database next click on continue button.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines (pipeline1 through pipeline4) and datasets (AzureSqlTable1, DelimitedText3 through DelimitedText8). The main workspace shows a pipeline diagram with a 'Lookup' activity connected to a 'ForEach' activity. The 'ForEach' activity is expanded, showing its configuration. On the right, the 'New dataset' wizard is open, with the 'Azure SQL' option selected. Below this, two options are shown: 'Azure SQL Database' and 'Azure SQL Database Managed instance'. The 'Continue' button is visible at the bottom of the wizard.

Step 11

In set properties make a new dataset name select linked service for Azure SQL database dont select any table next click on ok button.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists pipelines and datasets. The 'Activities' pane shows a 'Lookup' activity. The main canvas displays a 'Lookup' activity connected to a 'Foreach' loop. The 'Set properties' dialog box is open on the right, showing the configuration for the 'Lookup' activity. The 'Name' field is 'AzureSqlTable2'. The 'Linked service' is 'AzureSqlDatabase1'. The 'Table name' is 'Select...'. The 'Import schema' option is 'None'. The 'Source dataset' is 'Select...'. The 'OK' button is highlighted.

Step 12

Next click on pencil icon and open source dataset.

The screenshot displays the Microsoft Azure Data Factory portal interface. The left-hand navigation pane shows the 'Factory Resources' section, with 'pipelines' expanded and 'pipeline4' selected. The main workspace shows the 'Activities' tab for 'pipeline4', featuring a 'Lookup' activity named 'Lookup1' connected to a 'ForEach' activity named 'ForEach1'. The 'ForEach' activity is expanded, revealing its 'Activities' sub-tab. The 'Source' tab is active, showing the configuration for the 'Source dataset' as 'AzureSqlTable2'. The 'Use query' section has 'Table' selected. The 'Query timeout (minutes)' is set to '120'. The 'Partition option' is set to 'None'. The right-hand pane shows the 'Properties' tab for the selected activity, with the 'Name' field set to 'pipeline4'. The bottom status bar indicates the user is logged in as 'praveenads12 - Mic...' and the system time is 12:48 on 29-03-2025.

Step 13

Go to parameters create the following new parameters to hold schema name and table name.

The screenshot shows the Microsoft Azure Data Factory portal interface. The browser address bar displays the URL: `adf.azure.com/en/authoring/dataset/AzureSqlTable2?factory=%2Fsubscriptions%2Fbc1ab9c4-0eb1-4f48-98c1-8751cbcf738%2FresourceGroups...`. The page title is "Microsoft Azure | Data Factory | newadpraveendemo". The left sidebar shows the "Factory Resources" tree with categories like Pipelines (4), Datasets (8), and Data Flows (2). The main area displays the "pipeline4" configuration for the "AzureSqlTable2" dataset. The "Parameters" tab is active, showing a table with two parameters:

Name	Type	Default value
TABLE_SCHEMA	String	Value
TABLE_NAME	String	Value

The right sidebar shows the "Properties" panel for the selected parameter, with fields for "Name" (containing "AzureSqlTable2") and "Description". The bottom status bar shows the system clock as 12:50 on 29-03-2025.

Step 14

Now go to connections enable enter manually then go to schema and table name add dynamic content to set the table schema and table name.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' (4 items) and 'Datasets' (8 items). The 'AzureSqlTable2' pipeline is selected. The main workspace shows the 'Connection' tab for 'AzureSqlTable2', which is linked to 'AzureSqlDatabase1'. The 'Table' section has input fields for 'schema name' and 'table name', both containing dynamic content: '@dataset().TABLE_SCHEMA' and '@dataset().TABLE_NAME' respectively. A 'Test connection' button is visible. On the right, the 'Pipeline expression builder' pane is open, showing the dynamic content being added to the pipeline. The bottom of the screen shows the Windows taskbar with the search bar, system clock (12:50, 29-03-2025), and language (EN).

Step 15

Go back to copy activity source now go to table schema add dynamic content then go to ForEach iterator select ForEach1 current item put TABLE_SCHEMA after item(). Similarly , Go to TABLE_NAME add dynamic content do same thing for it but put TABLE_NAME after item().

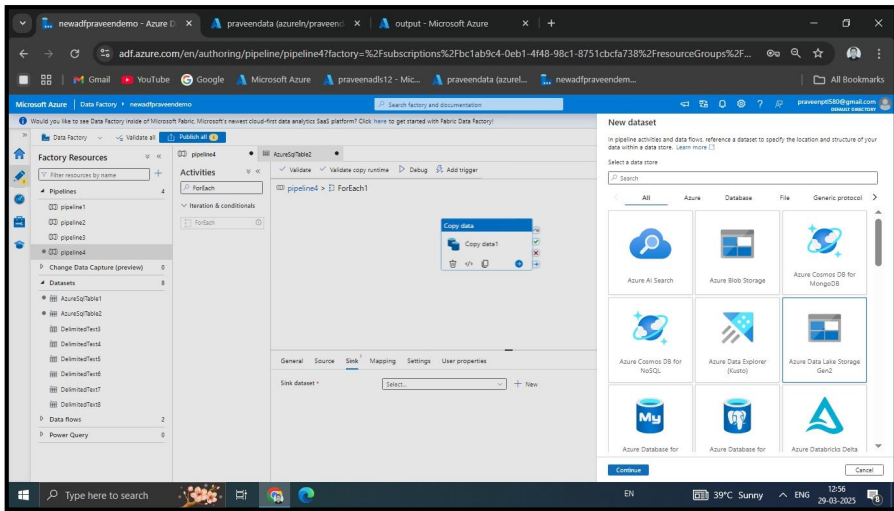
The screenshot displays the Microsoft Fabric Data Factory web interface. On the left, the 'Factory Resources' pane shows a hierarchy of pipelines and datasets. The 'Activities' pane for 'pipeline4' shows a 'ForEach' iterator. The main canvas shows a 'Copy' activity within a 'ForEach' iterator. The 'Source' tab is selected, showing the 'Source dataset' as 'AzureSqlTable2'. Below this, a table lists dataset properties:

Name	Value
TABLE_SCHEMA	Value Add dynamic content (Alt+Shift+Q)
TABLE_NAME	Value

At the bottom, there are radio buttons for 'Table' (selected), 'Query', and 'Stored procedure'. On the right, the 'Pipeline expression builder' pane is open, showing the expression '@item().TABLE_SCHEMA' in the 'TABLE_SCHEMA' field. Below this, there are tabs for 'ForEach iterator', 'Activity outputs', 'Parameters', and 'System variables'. The 'ForEach iterator' tab is active, showing 'ForEach1' and 'Current item'.

Step 16

Similarly go to copy activity sink create a new sink dataset search and select data store ADLS Gen2 for new sink dataset then click on continue. In dataset format delimited text then click on continue.



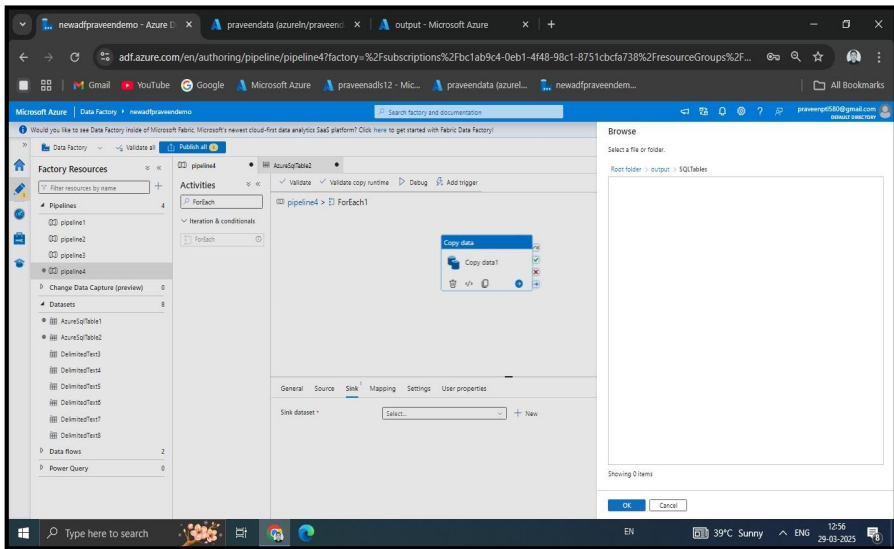
Step 17

In set properties select linked service for ADLS Gen2 next click on browse button to browse output container and SQLfiles folder to copy multiple files in bulk.

The screenshot shows the Microsoft Fabric Data Factory interface in a web browser. The 'Set properties' dialog for a 'Copy data' activity is open. The 'Name' field is set to 'DelimitedText8'. The 'Linked service' dropdown is set to 'AzureDataLakeStorage1'. The 'File path' section shows 'File system' selected, with 'Directory' and 'File name' fields. The 'First row as header' checkbox is checked. The 'Import schema' section has 'None' selected. The 'Sink dataset' dropdown is set to 'Select...'. The 'OK' button is highlighted.

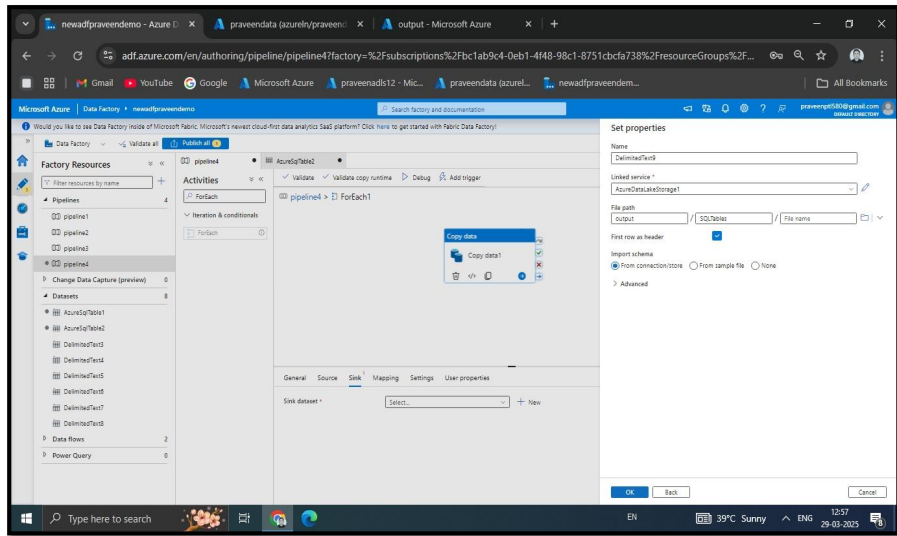
Step 18

Next go to output container then go to output SQLtables folder then click on ok.



Step 19

After output container and SQLtables folder are browsed successfully then finally click on ok button.



Step 20

Make sure give file extension as .csv if it is .txt or other.

The screenshot displays the Microsoft Azure Data Factory portal interface. The browser address bar shows the URL: `adf.azure.com/en/authoring/pipeline/pipeline4?factory=%2Fsubscriptions%2Fbc1ab9c4-0eb1-4f48-98c1-8751cbcf738%2FresourceGroups%2F...`. The page title is "Microsoft Azure | Data Factory | newadfravedemo".

The interface is divided into several sections:

- Factory Resources:** A sidebar on the left containing a search bar and a list of resources. Under "Pipelines", "pipeline4" is selected. Under "Datasets", "AzureSqlTable2" is selected.
- Activities:** A central pane showing the "pipeline4" activity. It contains a "ForEach" loop with a "Copy data" activity inside. A tooltip for the "Copy data" activity is visible, showing "Copy data1" and icons for file operations.
- Properties:** A right-hand pane showing the "Properties" tab for the selected activity. It includes fields for "Name" (set to "pipeline4") and "Description". Below this, there are tabs for "General", "Source", "Sink", "Mapping", "Settings", and "User properties". The "Sink" tab is currently active, showing configuration options for the data destination, including "Max concurrent connections", "Block size (MB)", "Metadata", "Quote all text", "File extension" (set to ".csv"), and "Max rows per file".

The Windows taskbar at the bottom shows the system clock as 12:57 on 29-03-2025, with a temperature of 39°C and weather conditions of Sunny.

Step 21

Go to parameters create the following new parameters to hold schema name and table name.

The screenshot shows the Microsoft Azure Data Factory portal interface. The left sidebar displays the 'Factory Resources' tree with 'Pipelines' (4 items) and 'Datasets' (9 items). Under 'Datasets', 'AzureSqlTable2' is expanded, showing a list of datasets including 'DelimitedText9', which is selected. The main pane shows the 'Parameters' tab for 'DelimitedText9'. A table lists the parameters:

Name	Type	Default value
TABLE_SCHEMA	String	Value
TABLE_NAME	String	Value

The right sidebar shows the 'Properties' panel for the selected dataset, with the 'Name' field set to 'DelimitedText9'.

Step 22

Next go to connections then go to File Path filename add dynamic content provide the following expression to concatenate scheme name with tablename and .csv extension in the expression builder.

The screenshot displays the Microsoft Azure Data Factory web portal. On the left, the 'Factory Resources' pane shows a list of pipelines (pipeline1 to pipeline4) and datasets (AzureSqlTable1, AzureSqlTable2, DelimitedText3 to DelimitedText9). The 'DelimitedText9' dataset is selected. The main pane shows the 'DelimitedText9' dataset configuration. The 'Connection' tab is active, showing 'AzureDataLakeStorage1' as the linked service. The 'File path' field is set to 'output / SQLTables'. The 'File name' field is highlighted, and the 'Add dynamic content' button is visible. The 'Pipeline expression builder' pane on the right is open, showing the expression: `@concat(dataset().TABLE_SCHEMA,".",dataset().TABLE_NAME,".csv")`. The 'Parameters' tab is selected, showing 'TABLE_NAME' and 'TABLE_SCHEMA' as available variables. The 'OK' button is visible at the bottom of the expression builder.

Step 23

Go back to copy activity sink now go to table schema add dynamic content then go to ForEach iterator select ForEach1 current item put TABLE_SCHEMA after item(). Similarly , Go to TABLE_NAME add dynamic content do same thing for it but put TABLE_NAME after item().

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline4' selected. The 'Activities' pane shows a 'ForEach' activity, and the 'Copy data1' activity is highlighted. The 'Sink' tab is active, showing the 'Sink dataset' as 'DelimitedText5'. The 'Dataset properties' table is visible, with columns for 'Name' and 'Value'. The 'TABLE_SCHEMA' and 'TABLE_NAME' rows are highlighted, and the 'Value' column for 'TABLE_SCHEMA' contains the expression 'item().TABLE_SCHEMA'. The 'Pipeline expression builder' pane on the right shows the expression 'item().TABLE_SCHEMA' entered. The 'Copy behavior' dropdown is set to 'Select...'. The bottom status bar shows the system clock as 13:02 on 29-03-2025.

Name	Value
TABLE_SCHEMA	item().TABLE_SCHEMA
TABLE_NAME	

Step 24

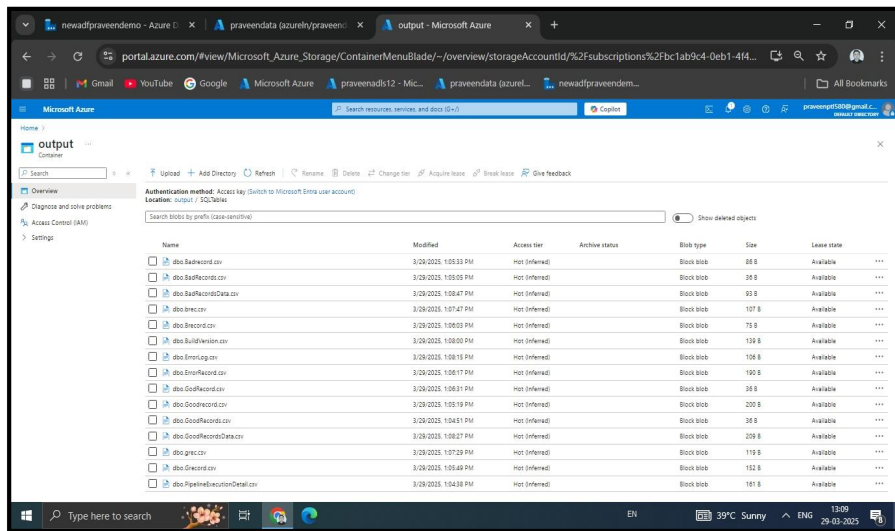
Now execute pipeline using debug mode when pipeline executes successfully it shows execution status as succeeded.

The screenshot displays the Microsoft Azure Data Factory portal interface. The top navigation bar shows the user is logged in as 'praveen@550@gmail.com'. The main content area is titled 'newadfpravedemo' and shows a pipeline named 'pipeline4' in the 'Activities' tab. The pipeline diagram shows a 'Lookup' activity followed by a 'ForEach' activity. The 'Pipeline status' is 'Succeeded'. Below the diagram, the 'Pipeline run ID' is '90697875-2f9e-4480-9071-a08b15150066'. The 'Pipeline status' is 'Succeeded'. The 'View debug run consumption' link is visible. The 'Output' tab shows a table of pipeline run details.

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	3/29/2025 1:09:30 PM	19s	AutoResolveIntegrationRuntime (East US)
Copy data1	Succeeded	Copy data	3/29/2025 1:08:17 PM	12s	AutoResolveIntegrationRuntime (East US)
Copy data1	Succeeded	Copy data	3/29/2025 1:08:03 PM	14s	AutoResolveIntegrationRuntime (East US)
Copy data1	Succeeded	Copy data	3/29/2025 1:07:49 PM	12s	AutoResolveIntegrationRuntime (East US)
Copy data1	Succeeded	Copy data	3/29/2025 1:07:34 PM	14s	AutoResolveIntegrationRuntime (East US)

Step 25

Go back to ADLS account then go to output container folder SQLTables when you click on refresh you will be able to see all the files that present in dbo schema.



Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
dbo.BadRecord.csv	3/29/2025, 1:05:33 PM	Hot (Inferred)		Block blob	86 B	Available
dbo.BadRecords.csv	3/29/2025, 1:05:05 PM	Hot (Inferred)		Block blob	39 B	Available
dbo.BadRecordsData.csv	3/29/2025, 1:08:47 PM	Hot (Inferred)		Block blob	93 B	Available
dbo.brec.csv	3/29/2025, 1:07:47 PM	Hot (Inferred)		Block blob	107 B	Available
dbo.Brecord.csv	3/29/2025, 1:06:03 PM	Hot (Inferred)		Block blob	75 B	Available
dbo.BuildVersion.csv	3/29/2025, 1:08:00 PM	Hot (Inferred)		Block blob	139 B	Available
dbo.ErrorLog.csv	3/29/2025, 1:08:15 PM	Hot (Inferred)		Block blob	108 B	Available
dbo.ErrorRecord.csv	3/29/2025, 1:06:17 PM	Hot (Inferred)		Block blob	190 B	Available
dbo.GoodRecord.csv	3/29/2025, 1:06:31 PM	Hot (Inferred)		Block blob	36 B	Available
dbo.GoodRecord.csv	3/29/2025, 1:05:19 PM	Hot (Inferred)		Block blob	200 B	Available
dbo.GoodRecords.csv	3/29/2025, 1:04:51 PM	Hot (Inferred)		Block blob	36 B	Available
dbo.GoodRecordsData.csv	3/29/2025, 1:08:27 PM	Hot (Inferred)		Block blob	209 B	Available
dbo.grec.csv	3/29/2025, 1:07:29 PM	Hot (Inferred)		Block blob	119 B	Available
dbo.Grecord.csv	3/29/2025, 1:05:49 PM	Hot (Inferred)		Block blob	152 B	Available
dbo.PipelineExecutionDetail.csv	3/29/2025, 1:04:38 PM	Hot (Inferred)		Block blob	181 B	Available



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